SMART-LAMP PROJECT

- Name : Zach Deering GitHub Link : https://github.com/zukery10/Smart-Lamp

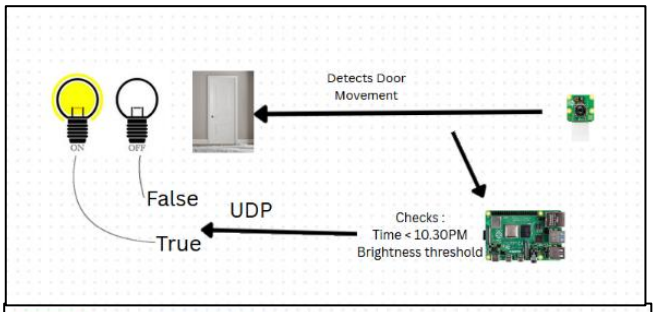
Student ID : 20119113

The smart lamp is an IoT project that utilises the Wiz smart bulb's ability to communicate using the protocol UDP. This project is an IoT based lighting and monitoring system that uses a Raspberry Pi, Sense HAT, Pi Camera (Picamera2) and as mentioned a Wiz Smart Bulb. The system creates a dynamic, adapting environment depending on temperature, brightness, time of day and motion. As well as this security features are also available if activated. All events that take place will be logged and a history will be made.

SYSTEM FEATURES:

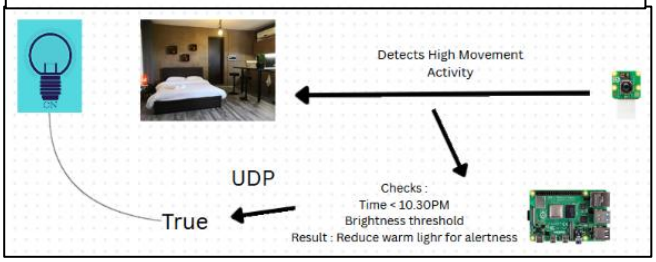
MOTION BASED LIGHTING

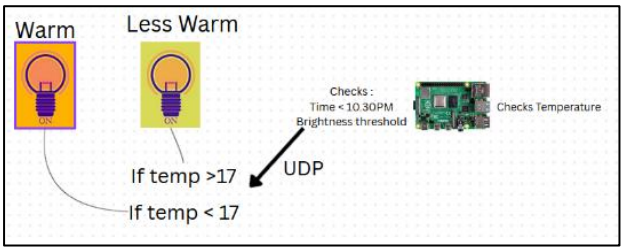
- The Pi Camera detects movement (door opening)
- Light activates if conditions are met, time is before 10.30pm and brightness threshold is not met



AMBIENT ADAPTIVE LIGHTING

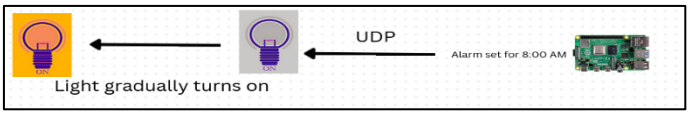
- High motion activity reduces warm light for alertness
- The Sense HAT temperature adjusts light warmth
  - Cooler temperature = Warmer Light
  - Warmer temperature = Cooler Light
- If PS5 is active on network = activate light **(if time permits) (Gaming mode)**
- Cooler light can also be programmed to inherit study/focus mode





WAKE UP LIGHTING

- User can activate alarm lighting to gradual light up in the mornings.



SECURITY MODE

- User can activate monitoring mode
- Motion detection alerts the user
- Motion detection takes a snapshot of motion

EVENT LOGGING AND ANALYTICS

- Motion detected, light on/off, temperature readings and security mode activated
  - Each event will be published using MQTT and logged to a file/web application. – with possible further integration with a home cloud storage system

